

- 4 (a) (i) *either* $\text{force} = e \times (V / d)$ *or* $E = V/d$ C1
 $= 1.6 \times 10^{-19} \times (250 / 7.6 \times 10^{-3})$ C1
 $= 5.3 \times 10^{-15} \text{ N}$ A1 [3]
- (ii) *either* $\Delta E_K = eV$ *or* $\Delta E_K = Fd$ C1
 $= 1.6 \times 10^{-19} \times 250$ $= 5.3 \times 10^{-15} \times 7.6 \times 10^{-3}$ M1
 $= 4.0 \times 10^{-17} \text{ J}$ A0 [2]
- (allow full credit for correct working via calculation of a and v)*

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(iii) either $\Delta E_K = \frac{1}{2}mv^2$
 $4.0 \times 10^{-17} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$ C1
 $v = 9.4 \times 10^6 \text{ m s}^{-1}$ A1 [2]

or $v^2 = 2as$ and $a = F/m$
 $v^2 = (2 \times 5.3 \times 10^{-15} \times 7.6 \times 10^{-3}) / (9.11 \times 10^{-31})$ (C1)
 $v = 9.4 \times 10^6 \text{ m s}^{-1}$ (A1)

- (b) speed depends on (electric) potential difference M2
(If states ΔE_K does not depend on uniformity of field, then award 1 mark, treated as an M mark)
 so speed always the same A1 [3]